

China Semiconductors

Montage: CXL deep dive - a new growth engine from DDR4 reuse



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As we highlighted in our [memory interface chip primer](#), the rollout of agentic AI is creating significant incremental demand for CPUs and thereby server DRAM. However, DRAM supply shortages have become a key bottleneck facing hyperscalers, as new solutions to increase supply became increasingly urgent. Meta recently published a [paper](#) using Compute Express Link (CXL) Memory eXpander Controller (MXC) to reuse retiring DDR4. Our discussion with Montage also confirmed that local CSPs are developing similar solutions. We updated our [company model](#) and [industry model](#) to reflect this upside. **We believe this could become a new growth engine for Montage, bring ~10% upside to their 2028 revenue, reiterate Outperform.**

We believe DDR4 reuse could become the first large-scale commercial application of CXL MXC. CXL has emerged in less than a decade as one of the most promising new interconnect technologies in AIDC. Recently, the market is increasing focus on a new use case on using CXL MCX for DDR4 reuse. A widening server-memory shortage, attractive reuse economics and growing AI memory footprints create a commercial entry point for the adoption of this technology. We estimate global CXL MXC TAM for DDR4 reuse could reach US\$8bn by 2030E. Assuming Montage captures 50% market share in China and 10% market share in non-China, we estimate incremental revenue potential equivalent to ~10% of our 2028E sales projection.

Memory shortages make retired DDR4 an attractive source of incremental capacity. We forecast global server DDR demand to grow at a 47% CAGR in 2025–28E, versus roughly 23.8% growth in overall DRAM bit capacity, with HBM competing for manufacturing resources. We estimate the global server DDR shortfall could reach ~5.5bn GB in 2027E and ~11bn GB in 2028E. Meanwhile, server replacement cycles of 5–7 years release DDR4 DIMMs with useful lives of roughly 10+ years. CXL enables DDR5-generation CPUs to access this memory through an MXC. Under our estimation, a 128GB reuse configuration costs ~US\$920, versus ~US\$3,500 for native DDR5, a 74% saving.

Performance trade-offs limit the addressable workloads, but AI is expanding the workloads that fit CXL's performance profile. CXL expansion offers roughly single-DDR4-channel-class bandwidth per x8 lanes connection, with low-load latency approximately 2–2.5x local DRAM and further deterioration at higher utilization. The strongest fit is capacity-intensive workloads with small hot working sets and limited sustained bandwidth demand. Agent sandboxes and VMs can reserve substantial memory while actively using only a portion. Inactive KV cache, long-tail RAG/vector data and less frequently accessed recommendation embeddings offer additional candidates. Effective tiering keeps hot data in local memory but warm pages in CXL DDR4.

Montage is well positioned to capture the initial reuse opportunity. An early MXC supplier and the only meaningful domestic merchant MXC vendor in China, Montage combines an established server-memory-interface customer base with a simple reuse architecture. Its M88MX5891 supports two DDR4 DIMMs through one channel and a PCIe 5.0 x8 connection, enabling smaller capacity increments at limited controller cost. General-purpose servers are the primary target given their greater available PCIe lanes. CSP qualification, card deployments and competitive share determine revenue conversion.

DDR4 reuse could lay the foundation for longer-term CXL growth trajectory. Beyond expansion, memory pooling can improve capacity utilization across hosts, while eventual memory sharing could reduce data duplication and movement across compute devices. We believe reuse deployments could accelerate hardware qualification and software maturity, lowering adoption barriers for these broader use cases and extending Montage's growth opportunity beyond the initial reuse cycle.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	21 Sep 2026	Price Target	TTM	Reported EPS				Reported P/E (x)		
			Closing Price		Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
6809.HK (Montage)	O	HKD	287.00	520.00	NA	CNY	1.97	3.44	5.69	124.3	71.2	43.1
OLD								3.24	5.59			
688008.CH (Montage)	O	CNY	210.10	400.00	50.5%	CNY	1.97	3.44	5.69	106.6	61.1	36.9
OLD								3.24	5.59			
ASIAx			1,953.49									

ESTIMATE CHANGE IN BOLD

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

Reiterate Outperform on Montage, PT HKD 520 for H-share and CNY 400 for A-share. We estimate the TAM for CXL MXC could reach \$8bn by 2028, bringing ~10% upside to Montage's revenue in 2028. We updated our [company model](#) and [industry model](#) to reflect this upside. EPS in 2027 and 2028 is raised by 2% and 11%, respectively. 2BF P/E ratio is cut to 47x for A-share. Maintain the target price.

EXHIBIT 1: **Bern forecast vs. Bloomberg Consensus vs. Bern old**

Revenue (RMB mn)	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Bernstein	2,286	3,639	5,456	7,059	14,030	27,936	47,739	69,520
Growth YoY	(38%)	59%	50%	29%	99%	99%	71%	46%
Consensus	2,286	3,639	5,456	7,597	11,484	16,096		
Growth YoY			50%	39%	51%	40%		
Bernstein vs. consensus				(7.1%)	22.2%	73.6%		
Bernstein old				7,324	13,662	25,322	42,075	58,408
New vs. Old				(3.6%)	2.7%	10.3%	13.5%	19.0%
Gross profit								
Bernstein	1,347	2,115	3,395	4,701	9,255	18,234	30,500	43,496
Growth YoY		57%	61%	38%	97%	97%	67%	43%
GP Margin	58.9%	58.1%	62.2%	66.6%	66.0%	65.3%	63.9%	62.6%
Consensus	1,347	2,115	3,395	5,012	7,928	11,626		
Growth YoY		57%	61%	48%	58%	47%		
GP Margin			62.2%	66.0%	69.0%	72.2%		
Bernstein vs. consensus - GP				(6.2%)	16.7%	56.8%		
Bernstein vs. consensus - GPM				62bps	(308bps)	(696bps)		
Bernstein old				4,884	8,996	16,386	26,544	35,839
New vs. Old				(3.7%)	2.9%	11.3%	14.9%	21.4%
Operating profit								
Bernstein	199	1,046	1,971	2,936	7,114	15,699	27,499	39,971
Growth YoY		426%	88%	49%	142%	121%	75%	45%
OP Margin	8.7%	28.8%	36.1%	41.6%	50.7%	56.2%	57.6%	57.5%
Consensus	199	1,046	1,971	3,401	5,694	8,358		
Growth YoY		426%	88%	73%	67%	47%		
OP Margin			36.1%	44.8%	49.6%	51.9%		
Bernstein vs. consensus - OP				(13.7%)	24.9%	87.8%		
Bernstein vs. consensus - OPM				(318bps)	112bps	427bps		
Bernstein old				3,118	6,857	13,865	23,573	32,371
New vs. Old				(5.8%)	3.7%	13.2%	16.7%	23.5%
Net income attributable to shareholders								
Bernstein	451	1,412	2,236	4,149	6,958	14,500	24,768	35,519
Growth YoY	(65%)	213%	58%	86%	68%	108%	71%	43%
NP Margin	19.7%	38.8%	41.0%	58.8%	49.6%	51.9%	51.9%	51.1%
Consensus	451	1,412	2,236	3,998	5,797	7,877		
Growth YoY			58%	79%	45%	36%		
NP Margin	19.7%	38.8%	41.0%	52.6%	50.5%	48.9%		
Bernstein vs. consensus - NP				3.8%	20.0%	84.1%		
Bernstein vs. consensus - NPM				615bps	(88bps)	297bps		
Bernstein old				4,311	6,732	12,894	21,347	28,948
New vs. Old				(4%)	3%	12%	16%	22.7%
Basic EPS								
Bernstein		1.25	1.97	3.44	5.69	11.74	20.03	28.44
Growth YoY			58%	75%	65%	106%	71%	42%
Consensus		1.25	1.97	3.25	5.13	6.82		
Growth YoY			58%	65%	58%	33%		
Bernstein vs. consensus				5.9%	10.9%	72.0%		
Bernstein old				3.24	5.59	10.60	17.54	23.18
New vs. Old				6%	2%	11%	14%	22.7%

Source: Bloomberg, Bernstein analysis and estimates

Table Of Contents

CXL: a short primer.....	5
The history of CXL.....	5
Three key use cases with CXL.....	6
Reshaping the datacenter memory hierarchy.....	8
Economics deep dive.....	11
\$8bn incremental CXL TAM by 2030 from DDR4 reuse.....	11
Global shortage on server DRAM.....	11
Elevated memory price makes the ROI attractive.....	14
Current available solution on CXL MXC.....	15
Tech feasibility deep dive.....	18
Sufficient PCIe channels to support CXL MXC.....	18
The solution comes with limitations on bandwidth and latency.....	19
AI is expanding the workloads that fit CXL DDR4 reuse.....	21

DETAILS

CXL: A SHORT PRIMER

THE HISTORY OF CXL

Compute Express Link (CXL) has emerged in less than a decade as one of the most consequential interconnect technologies in modern computing. Born from the industry's growing frustration with memory bottlenecks and stranded resources, CXL has evolved from a niche proposal into a universal standard that is fundamentally redesigning data center architecture. Its history traces a deliberate journey from point-to-point memory expansion to a full fabric for rack-scale composable infrastructure.

Origins: The Memory Wall and the Pre-CXL Landscape

By the late 2010s, the computing industry faced an intensifying "memory wall." CPU core counts were scaling rapidly, driven by Moore's Law and the rise of data-intensive workloads such as AI, in-memory databases, and cloud-native microservices. Yet memory capacity per server was constrained by physical DIMM slots and channel limits, and memory resources were often stranded—underutilized on one server while another nearby starved for capacity. Traditional PCIe, the dominant I/O interconnect, was not designed for cache-coherent memory sharing between processors and devices.

Several proprietary and semi-open approaches attempted to fill this gap, including IBM's OpenCAPI, Gen-Z, and Intel's Omni-Path. Each offered pieces of a solution but suffered from limited vendor buy-in, software incompatibility, or narrow use cases. The industry needed a single, open standard built on the ubiquitous PCIe physical layer that could deliver cache coherence, memory expansion, and pooling across heterogeneous compute environments.

Birth of the CXL Consortium and Version 1.0 (2019)

In March 2019, Intel formally announced Compute Express Link, and the CXL Consortium was incorporated shortly thereafter with founding members including Alibaba, Cisco, Dell EMC, Meta (then Facebook), Google, HPE, Huawei, Microsoft, and Intel. Crucially, rather than inventing an entirely new physical layer, CXL was built atop PCIe 5.0 electrical and physical infrastructure, leveraging the massive existing PCIe ecosystem and enabling reuse of existing PCB designs, connectors, and signal integrity knowledge.

CXL 1.0 and the follow-on 1.1 specification (released in June 2019) introduced three foundational protocols:

- **CXL.io**: Based on PCIe, for traditional I/O operations.
- **CXL.cache**: Allows accelerators to coherently cache host CPU memory.
- **CXL.mem**: Allows the host CPU to coherently access memory attached to a device.

This triad was revolutionary. For the first time, an industry-standard interconnect enabled bidirectional cache coherence between a host CPU and an endpoint device at near-native latencies.

The 2.0 Revolution: Pooling and Switching (2020)

The release of CXL 2.0 in November 2020 marked the technology's maturation from a point-to-point interconnect to a true fabric technology. The most significant addition was **CXL switching**, which enabled memory pooling across multiple hosts. Instead of each server owning its own dedicated memory pool, memory resources could be disaggregated into a shared pool and dynamically allocated to servers on demand.

CXL 2.0 also introduced support for persistent memory, enhanced RAS (Reliability, Availability, and Serviceability) features, and formalized the concept of memory expander controllers—later branded as MXC (Memory eXpander Controller) chips. This version arrived at a fortuitous moment: the AI boom was beginning, and hyperscalers were grappling with skyrocketing memory costs and supply constraints. DDR4 reuse via CXL memory expanders quickly emerged as a killer use case, allowing data centers to reuse millions of legacy DDR4 DIMMs from retiring servers into new DDR5-based AI clusters.

CXL 3.0 and Beyond: Fabric Composability (2022–Present)

Released in August 2022, CXL 3.0 doubled the per-lane bandwidth to 32 GT/s (matching PCIe 6.0) and introduced advanced fabric capabilities including multi-level switching, enhanced memory sharing across many hosts, and support for global shared memory spaces. The specification added flexible port configurations and improved Quality of Service (QoS), laying the groundwork for fully composable disaggregated infrastructure where compute, memory, and storage are independent resources allocated over a CXL fabric.

CXL 3.1, announced in 2024, further refined the standard with focus on AI-optimized features, lower latency modes, and improved management interfaces. By this point, the CXL Consortium had grown to over 300 member companies spanning semiconductors, systems, hyperscalers, and software vendors.

Industry Adoption and the Road Ahead

What began as an Intel-led initiative has become a truly cross-industry standard. AMD, Arm, and all major CPU vendors now support CXL on their server platforms. Marvell, Astera Labs, Montage, and others have launched commercial memory expander controller chips. Hyperscalers including Meta, Google, and Microsoft have publicly disclosed CXL deployments at scale—[Meta's Vistara ASIC](#), for instance, has recycled millions of DDR4 modules across its production fleet.

Looking forward, CXL is positioned to become the backbone of the AI-era data center. As agentic AI workloads drive insatiable demand for memory capacity and bandwidth, CXL's ability to disaggregate, pool, and share memory resources across racks will transition from a cost-saving optimization to an architectural necessity. In barely seven years, CXL has traveled from a technical specification to a transformative force—proof that when the industry unites behind an open standard, it can rewrite the rules of computing architecture.

DDR4 reuse could serve as the first killer application that moves CXL from ecosystem build-out into meaningful deployment, helping improve CSP familiarity, software maturity and hardware availability. Beyond simple expansion, memory pooling can reduce stranded capacity and improve utilization across CPU- and GPU-centric systems, while memory sharing could ultimately reduce data movement and improve system-wide utilization, giving CXL a much longer growth trajectory than DDR4 reuse alone.

THREE KEY USE CASES WITH CXL

CXL's three memory architectures address three distinct needs: expanding server memory capacity, improving memory utilization across servers, and enabling data sharing across compute nodes. Direct-attached expansion adds dedicated memory to an individual server; switch-based pooling allocates a common pool of DRAM capacity across multiple

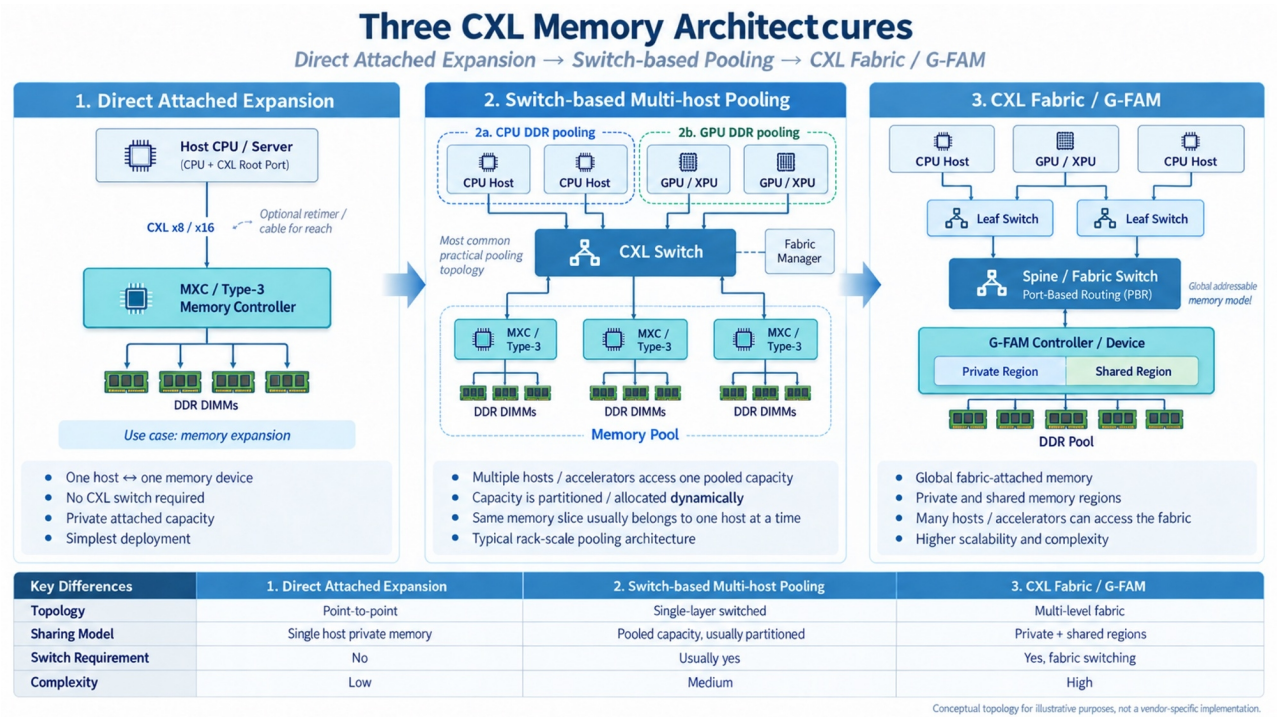
hosts; and CXL Fabric/G-FAM extends connectivity and supports shared memory across nodes. Hardware requirements increase accordingly, from a memory expansion controller to CXL switches and, ultimately, a multi-level fabric with support for cross-host memory coherency. The choice depends on workload requirements and the trade-offs between capacity, access performance, and system cost.

Memory expansion: Direct-attached expansion addresses the memory capacity constraints imposed by a server's CPU memory channels and DIMM slots. For memory-intensive workloads such as in-memory databases and data analytics, insufficient memory can force customers to add servers or upgrade platforms even when existing compute resources remain underutilized. In this architecture, a CXL-capable CPU root port connects over an x8/x16 PCIe lanes to a MXC, which in turn connects to DDR memory. The controller translates CXL.mem requests into accesses to the attached DDR, making the additional capacity addressable by the host. CXL switch is not mandatory, and the attached memory remains dedicated to that host. This provides a relatively simple expansion path, although application performance depends on link bandwidth, controller characteristics, and memory access patterns.

Memory pooling: Switch-based multi-host pooling addresses stranded memory capacity caused by uneven demand across servers. Provisioning each server for its own peak requirement can leave some nodes short of memory, while capacity sits idle elsewhere. A CXL switch connects multiple CPUs or GPUs to memory devices and their underlying DDR capacity, forming a common memory pool. A Fabric Manager assigns resources to hosts, with host software supporting capacity changes. This allows memory allocation to follow workload demand and can reduce excess provisioning. In typical CXL 2.0 pooling, each physical memory region belongs to only one host at a time: the benefit comes from reallocating capacity across hosts. The GPU/XPU pooling example similarly represents access to external DDR capacity by compatible accelerator systems; implementation requires appropriate accelerator and platform support and should not be interpreted as pooling GPU-local memory.

Memory sharing: CXL Fabric/G-FAM extends the architecture to larger systems and shared data access, with the potential to reduce duplicate memory copies and data movement across nodes. When multiple compute nodes use the same dataset, maintaining separate copies consumes additional capacity and creates synchronization traffic. The illustrated architecture connects CPUs, compatible GPUs/XPUs, and Global Fabric Attached Memory (G-FAM) devices through multiple layers of CXL switches, using Port-Based Routing (PBR) to support a larger fabric. The G-FAM controller manages the underlying DDR pool and exposes private and shared memory regions. With the required coherency support, multiple nodes can access the same shared region, supporting shared-data analytics and heterogeneous computing. However, commercial maturity remains a constraint. While vendors have introduced early memory-sharing solutions, the complete hardware and software ecosystem for large-scale, hardware-coherent sharing across a CXL fabric remains immature. Broader deployment requires coordinated support across host processors, switches, memory controllers, system software, and applications. We therefore expect this architecture to require a longer commercialization timeline than direct-attached expansion and conventional memory pooling.

EXHIBIT 2: **CXL's three memory architectures address three distinct needs: expanding server memory capacity, improving memory utilization across servers, and enabling data sharing across compute nodes**



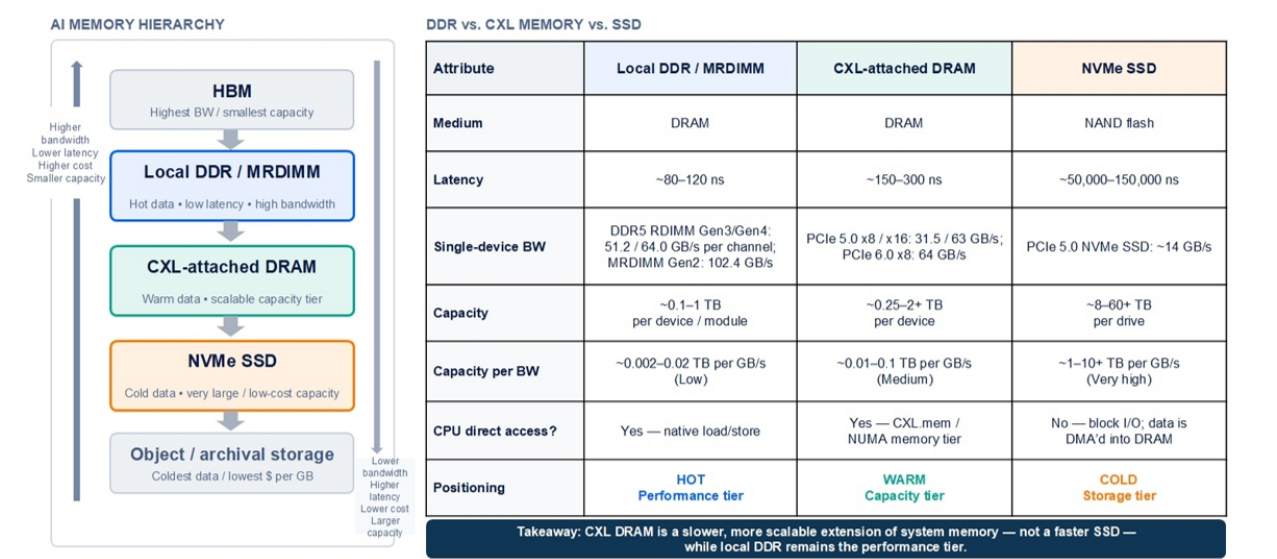
Source: CXL institution reports, Bernstein analysis

RESHAPING THE DATACENTER MEMORY HIERARCHY

Agentic AI strengthens the case for memory tiering, and CXL memory becomes an increasingly important layer.

Workloads place increasingly different demands on bandwidth, capacity, and latency, requiring more diverse memory segments. Model inference, retrieval, tool execution, and the retention of long-running agent context do not require every byte of data to reside in the fastest memory. With HBM capacity constrained by packaging and cost, and local DDR expansion limited by CPU memory channels and DIMM slots, a more granular hierarchy allocates scarce resources according to access frequency and performance requirements. HBM serves bandwidth-intensive accelerator workloads, local DDR holds frequently accessed CPU data, and NVMe SSDs and archival storage accommodate progressively colder data. CXL-attached DRAM adds a warm memory tier between local DDR and NVMe, delivering memory-level access latency far below NVMe while enabling more flexible capacity expansion beyond local DDR limits. As Exhibit 3 illustrates, CXL memory latency ranges at approximately 150–300ns, far faster than 50–150µs of NVMe SSDs, with roughly 0.25–2+TB per CXL device. PCIe 5.0 x8/x16 links provide theoretical one-way bandwidth ceilings of approximately 31.5/63GB/s before CXL protocol overhead. This positions CXL for less frequently accessed context, caches, and other capacity-intensive data that still benefit from memory-speed access, while software placement policies preserve faster tiers for performance-critical data.

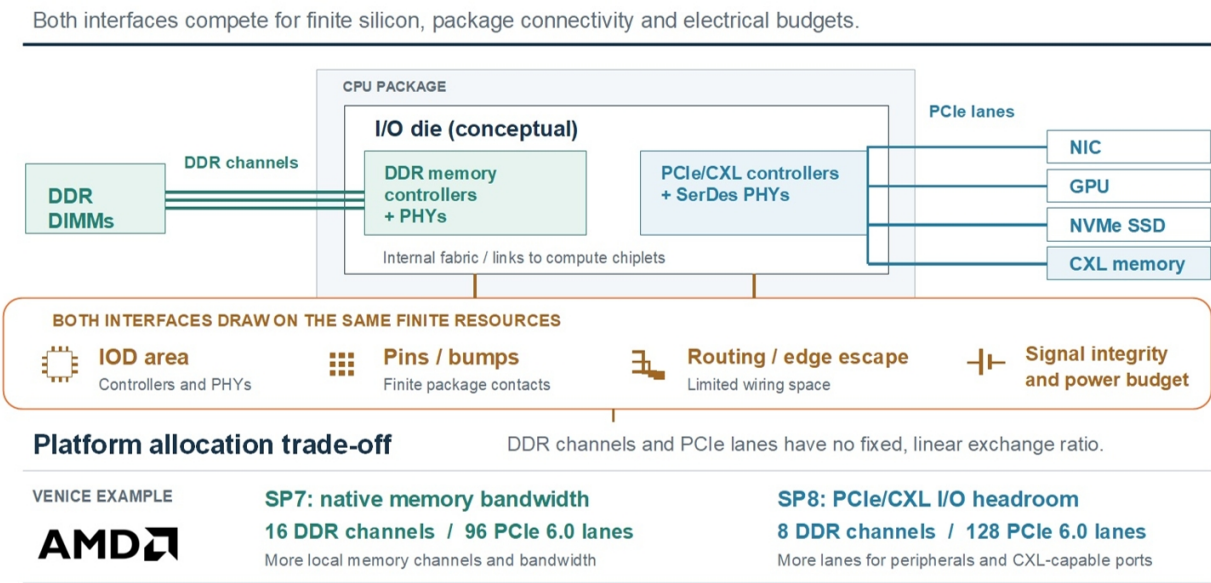
EXHIBIT 3: **CXL fills the gap between local DDR and NVMe by providing memory-speed access with flexible capacity expansion**



latency/ bandwidth ranges are implementation-dependent
Source: Meta Vistara report, companies reports, Bernstein analysis

CXL memory expansion is particularly relevant for CPU platforms with relatively few DDR channels and available CXL-capable I/O capacity, but its scalability remains constrained by finite platform resources. DDR controllers and PHYs compete with PCIe/CXL controllers and SerDes for silicon area, package pins and bumps, routing space, and power budgets. Within a given platform design budget, a higher DDR channel count can therefore leave less headroom for PCIe/CXL connectivity. CXL consequently cannot provide unlimited memory expansion: its capacity and bandwidth scale within the limits of available CXL-capable ports, lane allocation, and platform support. For CPUs with fewer DDR channels, connecting a Type-3 memory controller and its attached DRAM through available PCIe/CXL ports provides a practical way to increase addressable memory without adding native DDR channels. The trade-off is higher access latency and consumption of I/O lanes that could otherwise connect GPUs, NICs, or NVMe SSDs. We therefore view CXL's expansion value as platform-dependent: CPUs configured with more DDR channels provide greater native memory capability but may offer less PCIe/CXL headroom, while platforms with fewer DDR channels and more available CXL connectivity can use tiering to accommodate larger working sets.

EXHIBIT 4: **CXL offers a practical expansion path for CPUs with fewer DDR channels, but its scalability remains constrained by I/O resources. Shared I/O resources create a trade-off between PCIe/CXL and DDR**



Platform differences illustrate allocation choices, not a measured causal exchange of DDR channels for PCIe lanes
Source: AMD reports, Bernstein analysis

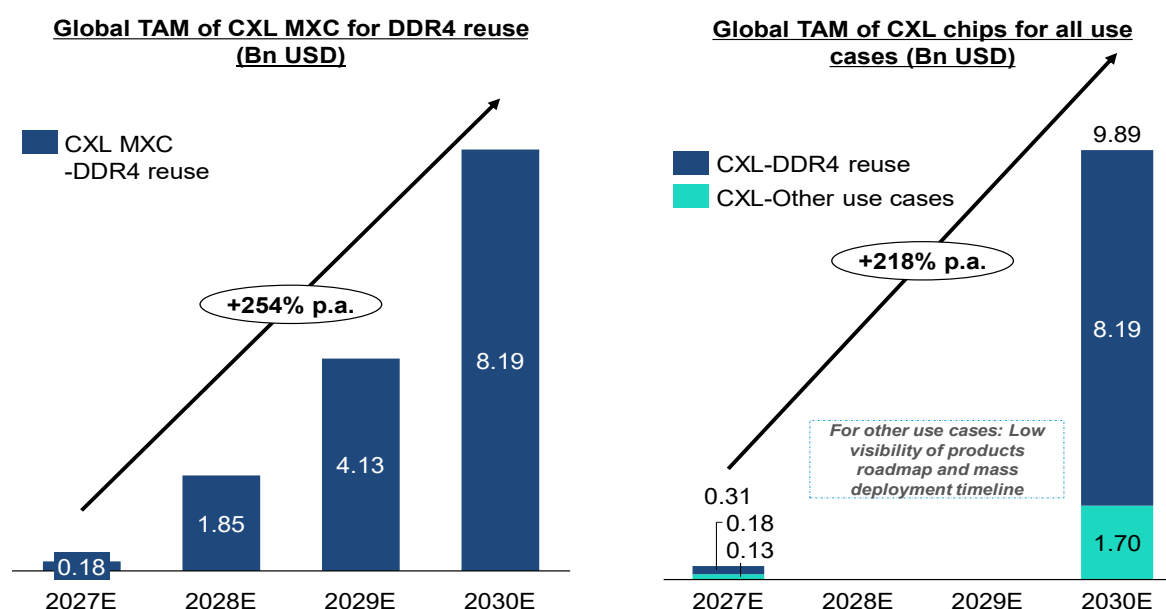
ECONOMICS DEEP DIVE

\$8BN INCREMENTAL CXL TAM BY 2030 FROM DDR4 REUSE

We believe DDR4 reuse could become the first large-scale commercial application of CXL memory expansion, creating a global MXC TAM of roughly US\$8.2bn by 2030E. Assuming Montage captures 50% of the China market and 10% of the non-China market, this could translate into roughly 10% potential revenue upside, making CXL a meaningful new growth driver beyond its core memory-interface business.

We estimate the TAM for CXL MXC used in DDR4 reuse through a bottom-up approach anchored in the server DDR supply-demand gap. We first estimate the shortfall in server DDR capacity and assume that CXL-enabled DDR4 reuse addresses 50% of this gap in 2028–2030. We then derive CXL MXC dollar content per GB of expanded memory capacity from existing controller solutions, based on controller pricing and the DDR4 capacity supported by each configuration. Multiplying the capacity shortfall addressed through DDR4 reuse by the implied MXC dollar content per GB yields our annual controller TAM.

EXHIBIT 5: We project DDR4 reuse will be the first killer application of CXL and its global TAM to reach 8.2Bn USD by 2030, driving CXL chips all use cases TAM to 9.9Bn USD in 2030

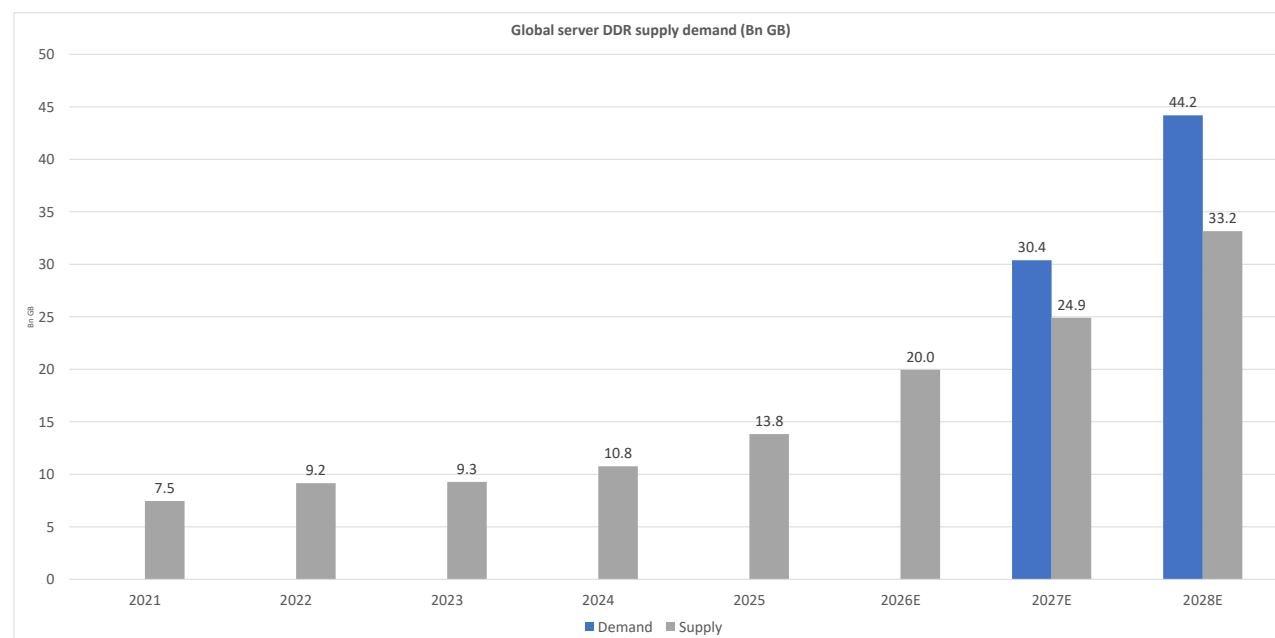


Source: Bernstein analysis and estimates

GLOBAL SHORTAGE ON SERVER DRAM

A structural server-memory shortage creates the need for DDR4 reuse, as server DRAM demand is growing much faster than the supply. We estimate the global server DDR shortfall will reach approximately 5.5bn GB in 2027E and 11bn GB in 2028E. These gaps establish the need for alternative capacity sources.

Our server DDR demand forecast implies a 47.3% CAGR over 2025–28E. Server DDR demand benefits from two reinforcing drivers: more CPUs and more memory attached to each CPU. We forecast server CPU shipments to grow at a 24% CAGR over 2025–30E, supported by the expansion of agentic AI. Over the same period, we forecast DDR capacity per CPU to grow at an 18% CAGR.

EXHIBIT 6: We project the global server DDR shortage in 2029 and 2030 to be 5.5Bn GB and 11Bn GB, respectively

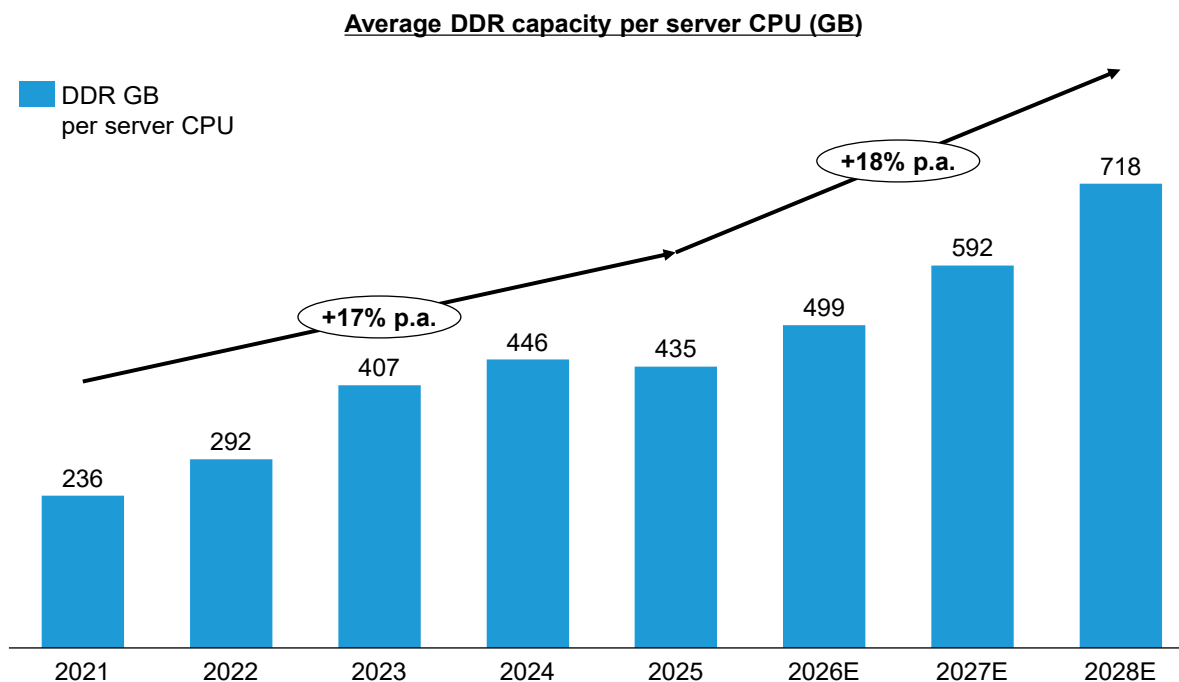
Source: TrendForce, Mercury, Bernstein analysis and estimates

Agentic AI extends infrastructure demand beyond model inference. Agents require environments in which to run browsers, execute code, retrieve information, call tools and maintain state across successive actions. These activities create CPU and memory requirements alongside accelerator demand. As more agents operate concurrently, the supporting infrastructure must accommodate more execution environments and larger aggregate application footprints.

Memory capacity per CPU grew at 17% CAGR during 2021-2025, and we expect the growth of the number will accelerate to 18% CAGR in the next two years. GB per CPU becomes increasingly important as operators consolidate workloads onto higher-core-count processors. Adding cores creates limited economic value if insufficient memory prevents those cores from hosting additional services. Larger working datasets, more concurrent sessions and retained application state increase the capacity needed to sustain CPU utilization. We therefore expect CPU DDR demand to grow faster than server CPU shipments alone.

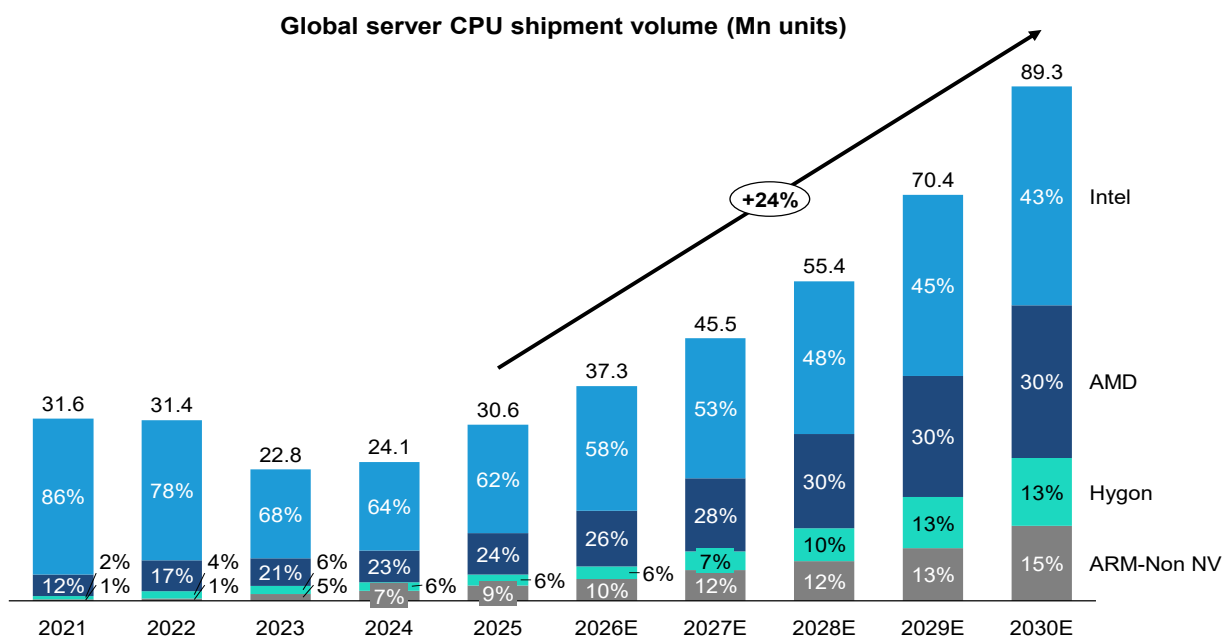
Bandwidth requirements increase alongside capacity, but the two do not rise uniformly across all data. Frequently accessed working sets require high bandwidth and low latency, while retained state and less active pages may consume substantial capacity with limited sustained traffic. This distinction creates room for a second memory tier: local DDR serves the hottest data, while a lower-cost tier accommodates a portion of the expanding footprint.

EXHIBIT 7: Memory capacity per CPU growth will accelerate, as more concurrent sessions and retained application state increase the capacity needed to sustain CPU and XPU utilization



Source: TrendForce, Mercury, Bernstein analysis and estimates

EXHIBIT 8: Due to rising demand of the Agentic AI, we forecast the server CPU shipment will grow at 24% CAGR during 2025 to 2030E



Source: Mercury, AMD and ARM reports, Bernstein analysis and estimates

Disciplined capacity expansion and HBM competition constrain server DDR supply. We expect DRAM supply growth to remain insufficient to meet the increase in server-memory requirements. We anticipate approximately 22% CAGR in

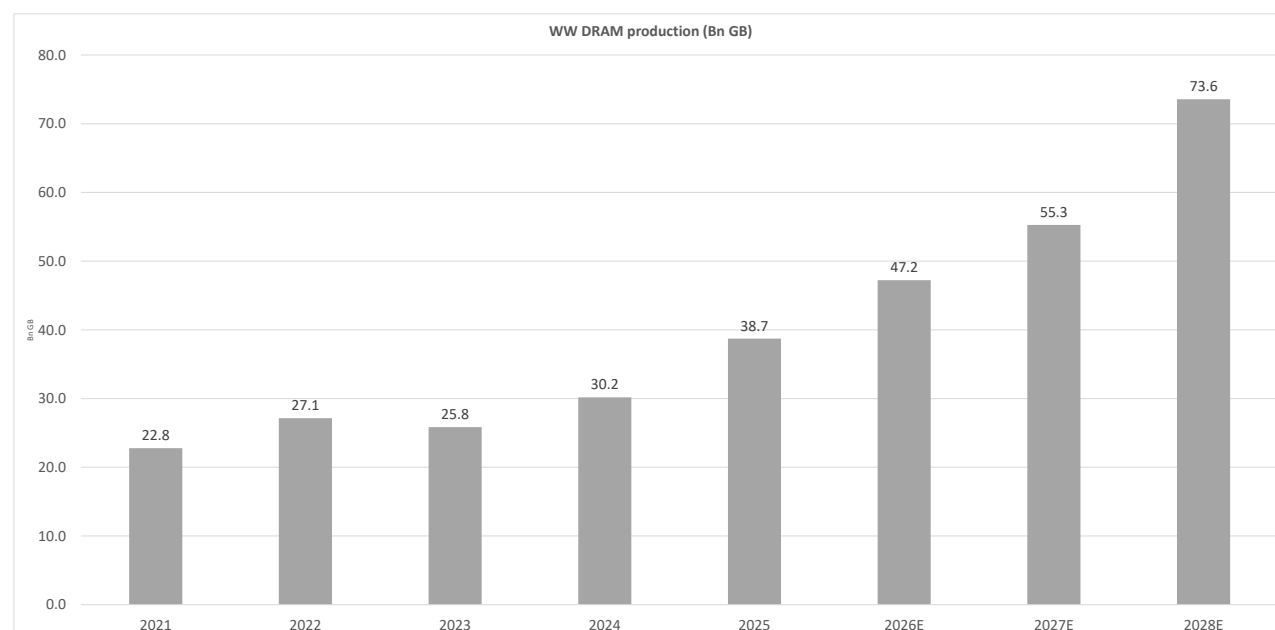
global DRAM bit capacity over 2026–28E, reflecting disciplined expansion and the time required to bring new capacity into qualified production. This is an aggregate DRAM measure; the availability of server DDR also depends on suppliers' allocation across products and end markets.

Announced investment does not immediately translate into usable memory supply. Suppliers must complete facilities, install equipment, ramp processes and qualify products before customers can deploy the resulting output.

HBM adds another constraint because it consumes manufacturing resources disproportionately to the number of bits produced. Micron has disclosed that HBM3E requires approximately three times the wafer supply of DDR5 to produce an equivalent number of bits on the same process node. As suppliers allocate more resources to HBM, growth in total investment need not generate a corresponding increase in conventional server DDR availability.

We also expect Chinese suppliers to increase output, but our forecast assumes that remaining technology and yield gaps versus global DRAM leaders limit their ability to eliminate the shortage. Wafer capacity alone is insufficient to assess the impact: qualified output, bit density, yields and the product mix determine how much additional supply can serve server demand. We therefore retain a supply-constrained outlook despite domestic expansion.

EXHIBIT 9: We anticipate the worldwide DRAM production will reach 73.6Bn GB by 2028, growing at the CAGR of 23.8% in 25-28E



Source: Bernstein analysis and estimates

ELEVATED MEMORY PRICE MAKES THE ROI ATTRACTIVE

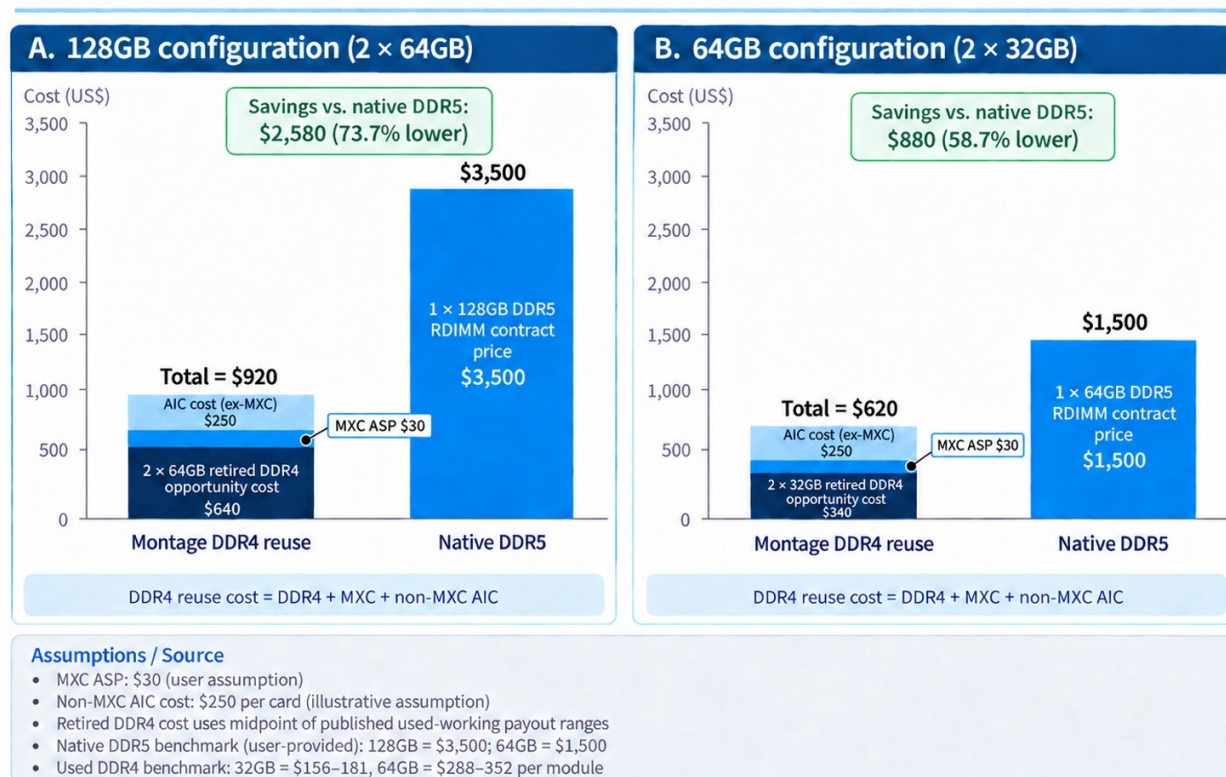
Server retirement can release usable DDR4 well before the memory reaches the end of its service life. In its July 2026 Vistara presentation, Meta cited a 5–7-year server retirement cycle versus a 10–14-year DRAM lifetime, illustrating the scope to reuse memory across successive server generations. These figures support the underlying economic case: platform retirement need not coincide with memory retirement. **We therefore see qualified DDR4 recovered from retired servers as a potential source of incremental capacity**, subject to module health, remaining service life and compatibility with the receiving platform.

CXL allows a DDR5-generation server CPU to access retired DDR4 through a memory-expander controller, effectively turning otherwise stranded memory into a usable warm-memory tier.

Based on current DDR5 contract pricing, we estimate that a CXL DDR4 reuse configuration can be 58–73% cheaper than same-capacity native DDR5; a 128GB configuration can save more than US\$2,580 even after including MXC and AIC costs. The DDR5–retired-DDR4 price spread can absorb the controller and card costs while leaving a substantial saving.

Our comparison uses August 2026 DDR5 contract-price benchmarks, per TrendForce, of US\$1,500 for a 64GB DIMM and US\$3,500 for a 128GB DIMM. We assume US\$30 per MXC and approximately US\$250 of non-MXC add-in-card costs. For a 128GB configuration, two reused 64GB DDR4 DIMMs carry an assumed aggregate opportunity cost of US\$640, the midpoint of used DDR4 price listed in SellUsedDRAM. Adding the controller and card brings the acquisition cost to US\$920, compared with US\$3,500 for equivalent native DDR5 capacity. This implies a saving of US\$2,580, or 74%. For 64GB, two reused 32GB DIMMs cost US\$340 on the same opportunity-cost basis; the US\$620 total reuse cost saves US\$880, or 59%, against native DDR5.

EXHIBIT 10: With Montage's solution as an illustrative example, DDR4 reuse via CXL can save more than half of cost, compared to native DDR5



used DDR4 costs use the midpoints of indicative buyback ranges of USD 156-181 per 32GB module and USD288-352 per 64GB. All memory prices reflect market condition in Aug 2026

Source: SellUsedDRAM, TrendForce, Bernstein analysis and estimates

CURRENT AVAILABLE SOLUTION ON CXL MXC

CXL MXC market is concentrated

The CXL MXC market is currently concentrated among a small group of suppliers, Montage, Astera Labs, Marvell and Meta's internal solution. The relevant competitive distinction is the architecture and economics of a qualified deployment. Their solutions cover smaller capacity increments, multi-channel expansion and a hyperscaler-specific design. Meta is a captive implementation rather than a merchant competitor.

Montage's M88MX5891 provides a PCIe 5.0 x8 path to DDR4 reuse, with more details discussed below.

Astera's newer Leo 2 E- and P-Series support PCIe 6.0, four DDR channels and configurations with twelve DDR4 DIMMs; Astera announced the expanded family in September 2026 and described it as sampling with hyperscaler customers. Its older CXL 2.0 Leo products primarily address DDR5 expansion.

Marvell's Structera X 2404 targets DDR4 expansion with 4 channels and up to 12 DIMMs, while the 2504 addresses DDR5. Using twelve 128GB DDR4 DIMMs provides 1,536GB of capacity. DIMM population also affects operating speed, as discussed

in the bandwidth section.

Meta provides direct evidence that DDR4 reuse can become an operational memory tier. Its July 2026 Vistara presentation describes a production platform with 768GB of local memory and 256GB of CXL memory, using two custom ASICs and Linux memory tiering. Each ASIC has two DDR4 channels with 2DPC; the production configuration uses four 32GB DIMMs per ASIC and x8 host connectivity. This validates the complete reuse approach.

EXHIBIT 11: Spec of current product offerings of CXL MXC for DDR4/DDR5 expansion

   				
1 DDR4-reuse-oriented MXC products			Targeting near-term cost efficiency by reusing existing DDR4 infrastructure	
Product	M88MX5891	Leo 2 / A2000 (DDR4 configuration)	Structera X 2404	Vistara (custom DDR4 reuse ASIC)
Host interface (PCIe / CXL, lanes, upstream BW)	PCIe 5.0 x8 / CXL 2.0 ~31.5 GB/s per direction	PCIe 6.0 x16 / CXL 3.2 128 GB/s raw per direction (Gen5 host: ~63 GB/s ceiling)	PCIe 5.0 x16 or 2 x x8 / CXL 2.0 ~63 GB/s per direction (x16)	PCIe 5.0 x16 / CXL 2.0 x8 in production: ~31.5 GB/s (per direction)
Memory topology (# channels, max DIMMs)	1 DDR4 channel 2 DPC, max 2 DIMMs	4 DDR4 channels 3 DPC, max 12 DIMMs	4 DDR4 channels Up to 3 DPC, max 12 DIMMs	2 DDR4 channels, 2 DPC 4 DIMMs per ASIC
Memory speed / downstream BW	DDR4-3200 25.6 GB/s (aggregate)	DDR4-3200 (up to 3 DPC) 4 x 25.6 = 102.4 GB/s (aggregate)	DDR4-3200, 2 DPC: 102.4 GB/s DDR4-1600, 3 DPC: 51.2 GB/s (aggregate downstream BW)	Production: DDR4-2400 2 x 19.2 = 38.4 GB/s (theoretical downstream BW / ASIC)
Total capacity	256GB configured capacity (using 2 x 128GB DIMMs)	1,536GB configured capacity (12 x 128GB DIMMs)	1,536GB configured capacity (12 x 128GB DIMMs)	Production: 128GB / ASIC (4 x 32GB; 256GB with 2 ASICs) Internal design; not merchant silicon
2 DDR5-expansion-oriented MXC products			Enabling higher capacity and bandwidth with DDR5 memory expansion	
Product	M88MX5851	Leo A1000 / first-gen Leo	Structera X 2504	No public merchant DDR5-expansion MXC
Host interface (PCIe / CXL, lanes, upstream BW)	PCIe 5.0 x8 / CXL 2.0 ~31.5 GB/s per direction	PCIe 5.0 x16 / CXL 2.0 ~63 GB/s per direction	PCIe 5.0 x16 or 2 x x8 / CXL 2.0 ~63 GB/s per direction (x16)	—
Memory topology (# channels, max DIMMs)	Not publicly disclosed	2 DDR5 channels 4 DDR5 RDIMM slots total	4 DDR5 channels 2 DPC, max 8 DIMMs	—
Memory speed / downstream BW	Up to DDR5-5600 (channel count / aggregate BW not publicly disclosed)	DDR5-5600 2 x 44.8 = 89.6 GB/s (aggregate)	DDR5-6400 4 x 51.2 = 204.8 GB/s (aggregate)	—
Total capacity	Not publicly disclosed	Up to 2TB	>6TB-class (DDR5 support)	—

configured capacity is not a controller maximum, but is estimated on mainstream DDR4 DIMMs capacity

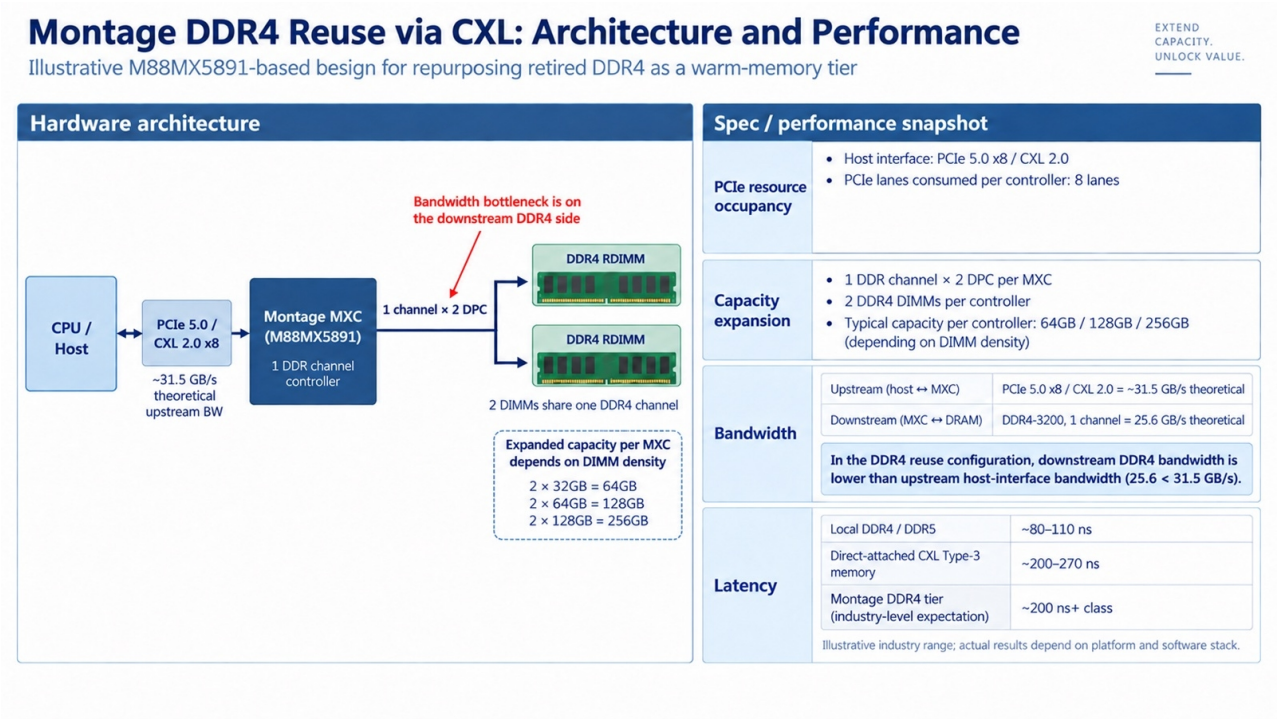
Source: companies reports, Bernstein analysis

Montage offers a simple, low-cost architecture well suited to DDR4 reuse

Montage's M88MX5891 uses a relatively simple PCIe 5.0 x8 + single DDR4 channel + 2DPC architecture, making it particularly suitable for lower-cost DDR4 reuse and positioning Montage as the most direct domestic beneficiary of adoption in China. Montage's opportunity is to make a useful capacity increment economical without requiring a large expansion subsystem. This provides a smaller deployment unit than the 12-DIMM designs by ALAB and Marvell, and can fit workloads that require incremental warm capacity rather than terabytes of additional memory.

Montage's existing memory-interface business provides relevant relationships across the server-memory supply chain. We believe this can support CXL MXC evaluation and integration, particularly in China. However, CXL qualification introduces additional requirements around host interoperability, firmware, error handling and software placement, requiring more time for large-scale deployment.

EXHIBIT 12: **Montage's current CXL MXC for DDR4 reuse, M88MX5891, uses a relatively simple PCIe 5.0 x8 + single DDR4 channel + 2DPC architecture, making it particularly suitable for lower-cost DDR4 reuse**



Source: company reports, Bernstein analysis

TECH FEASIBILITY DEEP DIVE

SUFFICIENT PCIE CHANNELS TO SUPPORT CXL MXC

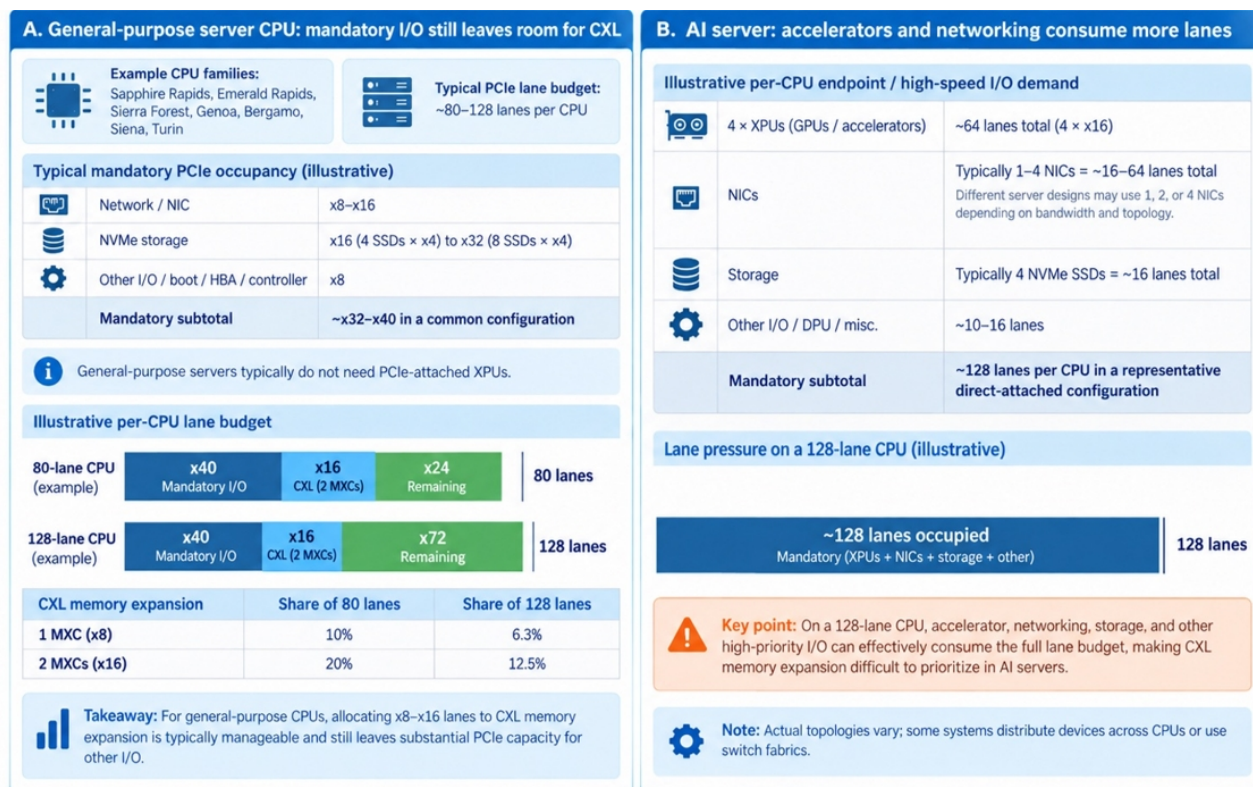
Our analysis shows that PCIe lane availability is generally sufficient on general-purpose CPUs. Unlike AI servers where accelerators and high-speed networking consume most of the available PCIe budget, allocating x8–x16 lanes to one or two MXCs is generally manageable for general-purpose server CPUs with roughly 80–128 PCIe lanes.

General-purpose servers provide a more accessible initial deployment environment because their PCIe resources face less competition from accelerators. Representative platforms include Intel Sapphire Rapids, with 80 PCIe 5.0 lanes per CPU, and AMD EPYC 9004 in a single-socket configuration, with 128 lanes, of which up to 64 can support CXL connectivity. These examples establish a useful range for evaluating the incremental resources required by one or two MXCs. In our illustrative general-purpose configuration, networking, NVMe storage and other required I/O consume 40 lanes. An 80-lane CPU retains 40 lanes before CXL, while a 128-lane CPU retains 88. Adding two x8 lanes MXCs consumes 16 lanes, leaving 24 and 72 lanes, respectively. The precise allocation varies by server, but these examples show why modest CXL expansion can fit within a conventional I/O budget. Available lanes must also connect to usable CXL-capable ports and slots. Motherboard routing, supported device counts, firmware and platform qualification determine whether nominal headroom translates into deployable memory. We therefore assess the complete server configuration rather than assuming every unallocated PCIe lane is interchangeable.

Accelerator-heavy servers face a higher opportunity cost. GPUs, high-speed NICs and storage compete for connectivity and bandwidth, with PCIe switches often shaping the final topology. CXL expansion may still be technically possible, but an additional MXC competes with resources that directly support accelerator utilization.

Exhibit 5. General-purpose servers retain more PCIe headroom for one or two MXCs

EXHIBIT 13: General-purpose servers typically have spare PCIe lanes for 1-2 CXL MXCs, while AI servers allocate much more high-speed I/O to XPU and NICs



Illustrative lane budgets for conceptual comparison. Exact lane usage depends on server topology, CPU generation, NIC count, storage design, and XPU attachment scheme

Source: Intel and AMD products spec, Bernstein analysis and estimates

THE SOLUTION COMES WITH LIMITATIONS ON BANDWIDTH AND LATENCY

Bandwidth is comparable to one native DDR channel, not the full local-memory subsystem

CXL expansion bandwidth depends on the complete memory path, not simply the number of DIMMs attached. A PCIe 5.0 x8 link provides approximately 31.5GB/s per direction before transaction-level overhead, while x16 provides approximately 63GB/s. A single DDR4-3200 channel provides 25.6GB/s of theoretical data bandwidth. Sustainable application bandwidth is lower and depends on access patterns, controller efficiency and protocol overhead.

Two bottleneck scenarios follow:

- In a multi-channel configuration whose combined DDR bandwidth exceeds the host link's capacity, the upstream CXL connection limits throughput. Additional DIMMs can expand capacity substantially without increasing host-visible bandwidth proportionately.
- In a single-channel DDR4 configuration such as the Montage design used here, the downstream 25.6GB/s DDR4 interface provides the lower theoretical ceiling relative to the x8 host link.
- DIMM population can change this comparison. Marvell's Structera X2404 supports DDR4-3200 at 2DPC, but its product brief specifies DDR4-1600 at 3DPC. Across four channels, these configurations imply theoretical downstream bandwidth of 102.4GB/s with eight DIMMs and 51.2GB/s with twelve DIMMs, respectively. The twelve-DIMM configuration therefore cannot automatically be treated as host-link-limited solely because it has four DDR channels.

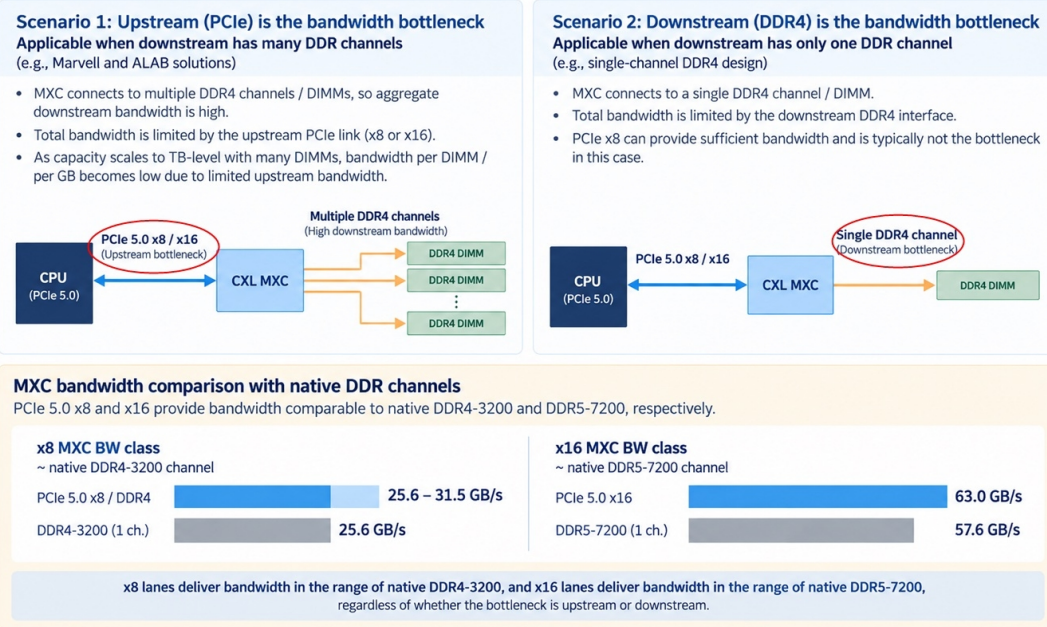
The appropriate comparison for a single Montage DDR4 expander is approximately one native DDR4-3200 channel. A sufficiently provisioned PCIe 5.0 x16 expander has a host-link ceiling in the same broad range as a fast native DDR5 channel, although actual throughput remains configuration-dependent. Neither comparison equates an individual expander with the CPU's complete multi-channel local-memory subsystem.

CXL can still add aggregate system bandwidth when it supplements existing local memory. The performance trade-off arises when data that would otherwise receive local-memory service moves to the slower tier, or when many workloads concentrate requests on one expander. Deployment should therefore preserve sufficient local capacity and bandwidth for the active working set.

EXHIBIT 14: Total bandwidth is limited either the upstream PCIe link or the downstream DDR4 interface, depending on the design

CXL MXC DDR4 reuse: bandwidth performance penalty

Total bandwidth is limited by either the upstream PCIe link or the downstream DDR4 interface, depending on the design.



Source: Companies products brief, Bernstein analysis and estimates

Latency is the more meaningful performance penalty

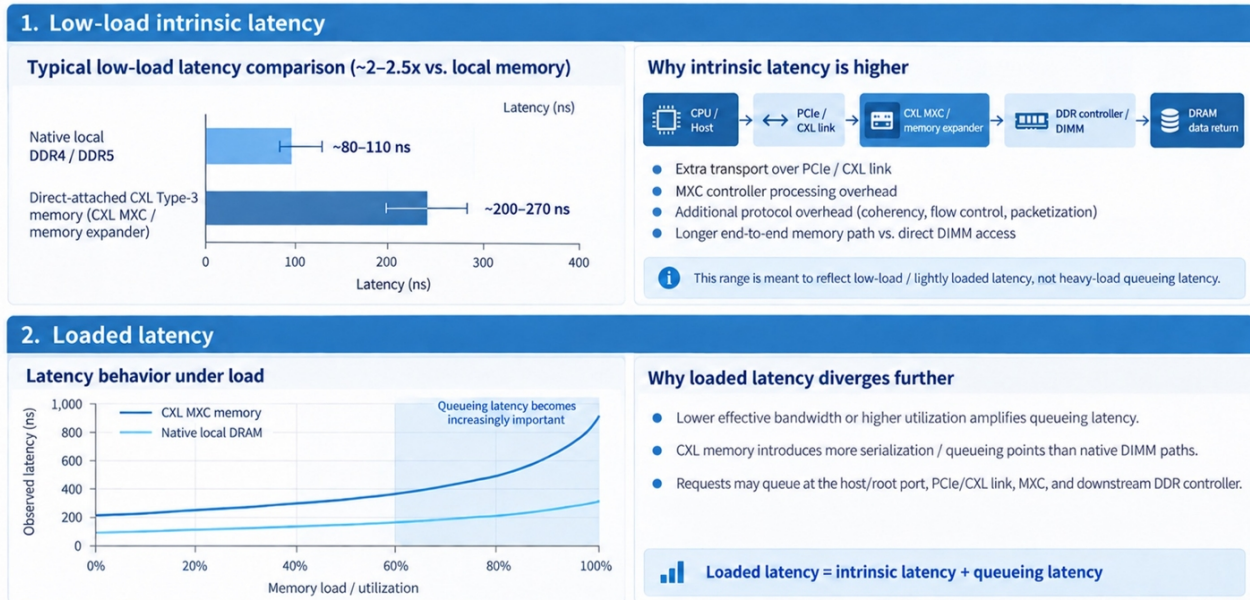
Latency is the more consequential constraint for workloads that repeatedly access data on the critical path. We use approximately 200–270ns as an illustrative low-load range for selected direct-attached CXL ASIC implementations, **broadly around 2–2.5x local DRAM in comparisons.** Published characterization reports ASIC-device latencies of 214ns, 239ns and 271ns, while also showing substantial variation across platforms and implementations. These measurements are reference points rather than a guaranteed Montage specification. CXL performance characterization.

The extra delay reflects the longer access path. A request traverses the host interface, CXL transport and expander before reaching the DDR controller and DIMM. Controller design and physical implementation affect the result, so the protocol alone does not determine a universal latency penalty.

Under load, queuing adds to this baseline. Requests can accumulate when traffic approaches the service capacity of the host link or downstream DDR interface. Shared links, controller queues and concurrent access streams influence how quickly latency rises.

The economic implication is that capacity utilization and bandwidth utilization should be assessed separately; hence, we do not expect full unsatisfied server DDR demand will be penetrated by the CXL DDR4 reuse solution. A card can be valuable while storing a large amount of infrequently accessed data and carrying modest traffic. Conversely, a smaller dataset can perform poorly, if it generates intense random accesses or sustained transfers. Customers must validate throughput and tail latency at the intended concurrency, including migration traffic between tiers.

EXHIBIT 15: CXL MXC memory expansion faces latency performance penalty, with intrinsic latency in low-load scenario and additional queuing latency in heavy-loaded scenario



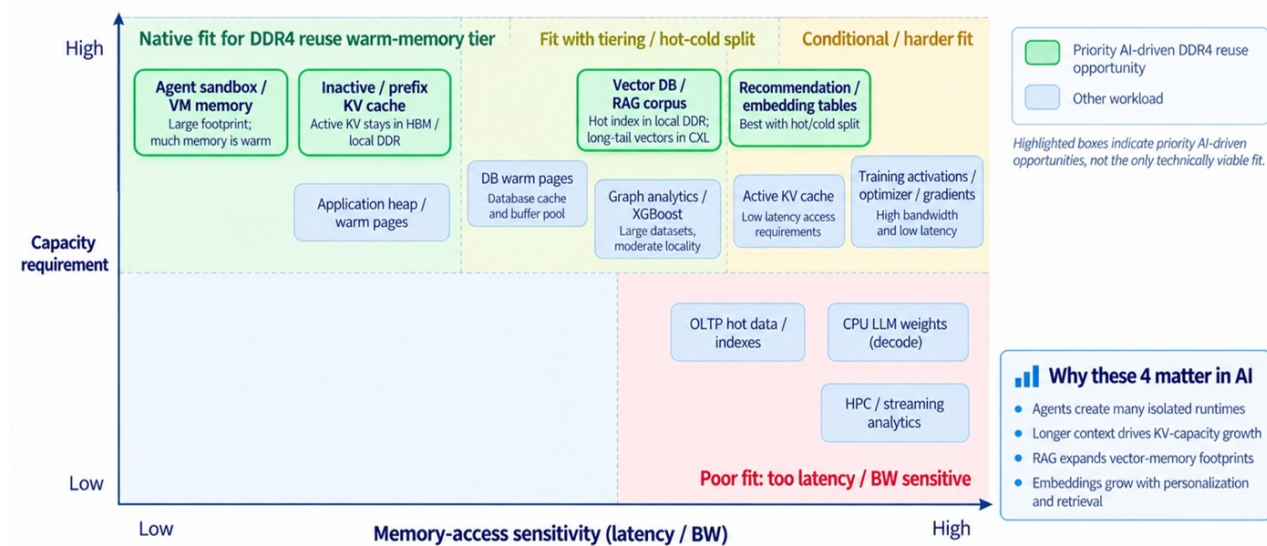
illustrative industry-level ranges and conceptual load curves; exact latency depends on platform, controller design, topology, utilization, and workload
Source: Microsoft report, Bernstein analysis

AI IS EXPANDING THE WORKLOADS THAT FIT CXL DDR4 REUSE

The best-fit workloads require large capacity but only a limited hot working set. The right way to assess CXL DDR4 reuse is not simply whether a workload is memory-intensive, but whether it needs large capacity without requiring all data to be accessed at native-DRAM latency and bandwidth. The strongest reuse candidates require substantial resident capacity while concentrating most accesses on a smaller hot working set. Capacity determines the potential procurement saving; the traffic directed to the slower tier determines the performance cost. We therefore assess workloads along two dimensions: memory-capacity requirements and sensitivity to memory access, illustrated by Exhibit 16. The distinction applies within applications as well as across them. A database may have latency-critical indexes alongside less active buffer pages. An inference service may hold active KV blocks alongside reusable state from inactive sessions.

Production offloading experience supports this approach. Meta's Transparent Memory Offloading work reported 20–32% memory savings across its deployment by identifying colder pages and moving them to lower-cost resources. That work supports the existence of tiering opportunities. For DDR4 reuse, the useful middle ground is data that benefits from remaining memory-addressable but does not need continuous local-DRAM performance. Data with virtually no reuse may fit better on storage, while frequently accessed data should retain the fastest appropriate tier. The size of this middle ground determines the useful capacity available to CXL.

EXHIBIT 16: We assess workloads by two dimensions: memory-capacity requirements and sensitivity to memory access. The best-fit workloads for CXL DDR4 reuse require large capacity but limited hot working set



conceptual positioning

Source: company reports, Bernstein analysis

Agent sandbox and VM memory offer a compelling use case of CXL DDR4 reuse

We see agent sandbox and VM memory as a particularly compelling incremental opportunity because concurrency can expand allocated capacity faster than the active working set. Agents may require isolated browsers, Python environments, application processes and retained session state. Many such environments can remain resident while awaiting model responses, external tools or user input, creating periods during which portions of their memory receive limited traffic.

This creates a potential separation between the memory required to preserve an execution environment and the memory required to advance its current task. Warm pages can remain in CXL DDR4 while actively used code and data stay in local DDR. If the platform can preserve application responsiveness, operators may support more concurrent environments without purchasing equivalent additional native DDR5 capacity. We expect the strongest adoption case where environments have a substantial warm footprint, activity is staggered and performance targets allow measured tiering. This places agent infrastructure naturally within the general-purpose server opportunity discussed above.

RAG/vector DB and embedding tables provide additional opportunities

Inactive or reusable KV cache creates a capacity opportunity when retaining state costs less than recomputing or retrieving it. Longer contexts and concurrent sessions increase the volume of potentially reusable KV data. Blocks associated with paused sessions or cached prefixes may remain valuable without being accessed continuously. CXL DDR4 reuse focuses on inactive or reusable blocks that can be promoted or transferred before reuse. The economic benefit depends on whether retention avoids sufficient recomputation and whether transfer time fits the application's response target.

RAG and vector databases offer a related opportunity in their less frequently accessed data. A growing knowledge base increases the number of vectors and associated records, while query popularity can concentrate accesses on a subset. Hot search structures and frequently accessed vectors can remain local, with an appropriate long tail assigned to CXL. Astera's published RAG discussion describes this separation between hot vectors in local DRAM and warmer data in CXL memory. The qualification is that retrieval itself can be latency-sensitive. Graph traversal, pointer chasing and concurrent queries may expose slower accesses directly to users. A large corpus creates a capacity opportunity, but its suitability for DDR4 reuse depends on the index design, access distribution and achieved tail latency. We therefore treat RAG as a tiering opportunity rather than assuming the entire vector database belongs in CXL.

Recommendation and embedding tables are attractive when accesses are sufficiently concentrated. Large tables can consume substantial memory while a smaller set of entries accounts for much of the traffic. Keeping frequently accessed embeddings local and assigning colder entries to a capacity tier can reduce native-memory requirements. However, random

lookups remain performance-sensitive, so the cold fraction must be large enough and sufficiently stable to justify placement. Published AMD–Asteria work demonstrates CXL's relevance to recommendation models, though its configuration uses DDR5 expansion.

EXHIBIT 17: **Four AI workloads offer different paths to monetize a warm-memory tier, fit with CXL DDR4 reuse**

AI workload	Why AI demand grows	Hot-set ratio	Acceptable latency	Typical memory footprint	Required BW	DDR4 reuse fit	Implementation note
Agent sandbox / VM memory	More concurrent agents need isolated runtimes, tools, file systems, and working environments	Low; much of allocated memory is idle or warm	Medium-high for warm pages	Tens to hundreds of GB per host; fleet-level TBs	Low to medium; bursty	Very high	Keep active runtime state in local DDR; place warm pages in CXL DDR4 tier
Inactive / prefix KV cache	Longer context windows and more concurrent sessions expand KV-cache capacity needs	Low-medium; only active KV is truly hot	Medium for inactive or reusable KV; low for active KV	Hundreds of GB to multi-TB per cluster	Low to medium for inactive tier	High	HBM/local DDR for active KV; CXL DDR4 for inactive, reusable, or prefetched KV blocks
Vector DB / RAG corpus	Enterprise RAG and retrieval systems drive very large embedding corpora	Medium-low; hot index small, long tail much larger	Medium; retrieval path is sensitive but tail data can be slower	Hundreds of GB to TB+	Medium; random access	Medium-high with tiering	Keep hot index and top-hit vectors in local DDR; place long-tail corpus in CXL DDR4
Recommendation / embedding tables	Personalization and retrieval use ever-larger embedding tables	Medium-low due to skew; small hot set, large long tail	Medium-low; sensitive for hot lookups	Hundreds of GB to multi-TB	Medium-high; random lookups	Medium with hot/cold split	Retain hot embeddings in local DDR; use CXL DDR4 for long-tail tables

 Key takeaway

- Best fit = large memory footprint, low hot-data ratio
- CXL DDR4 is a warm-memory tier, not a DDR5 replacement
- AI creates new demand in agents, KV cache, RAG, and embeddings

illustrative workload characteristics; actual thresholds also depend on software stack and hot-set management

Source: Bernstein analysis

APPENDIX - FINANCIAL FORECASTS

EXHIBIT 18: Montage income statement

INCOME STATEMENT	FY 2021	FY 2022	FY 2023	FY 2024	FY 2025	FY 2026	FY 2027	FY 2028	FY 2029	FY 2030
Revenue	2,562	3,672	2,286	3,639	5,456	7,059	14,030	27,936	47,739	69,520
Revenue YoY%		43%	(38%)	59%	50%	29%	99%	99%	71%	46%
Revenue QoQ%										
Quarterly revenue mix										
Cost of sales (enter as -)	(1,330)	(1,967)	(939)	(1,524)	(2,061)	(2,358)	(4,775)	(9,702)	(17,239)	(26,024)
Gross Profit	1,232	1,706	1,347	2,115	3,395	4,701	9,255	18,234	30,500	43,496
Gross profit margin%	48.1%	46.4%	58.91%	58.13%	62.2%	66.6%	66.0%	65.3%	63.9%	62.6%
YoY Bp Δ		-164 bps	1247 bps	-78 bps	410 bps	437 bps	-63 bps	-69 bps	-138 bps	-132 bps
Other Operating Income	76	1	2	12	116	43	71	71	71	71
Selling Expenses	(45)	(86)	(90)	(96)	(118)	(137)	(160)	(181)	(199)	(216)
General & admin expenses	(198)	(208)	(179)	(203)	(480)	(470)	(531)	(584)	(637)	(688)
Total SG&A	(243)	(295)	(269)	(299)	(598)	(606)	(691)	(765)	(836)	(904)
Research and development expenses	(370)	(563)	(682)	(763)	(915)	(1,139)	(1,378)	(1,626)	(1,919)	(2,264)
Other Operating Expense	(42)	(356)	(199)	(19)	(28)	(63)	(143)	(215)	(317)	(429)
Total operating expenses	(655)	(1,214)	(1,150)	(1,081)	(1,541)	(1,808)	(2,212)	(2,605)	(3,071)	(3,597)
Expenses analysis & assumptions										
Selling Expense YoY growth		92%	4%	7%	23%	16%	17%	13%	10%	8%
Admin Expense YoY growth		5%	(14%)	13%	137%	(2%)	13%	10%	9%	8%
SG&A Ratio	(9%)	(8%)	(12%)	(8%)	(11%)	(9%)	(5%)	(3%)	(2%)	(1%)
R&D YoY growth		52%	21%	12%	20%	24%	21%	18%	18%	18%
R&D Ratio	(14%)	(15%)	(30%)	(21%)	(17%)	(16%)	(10%)	(6%)	(4%)	(3%)
Financial income/ (expenses) as % of b		2%	3%	4%	4%	6%	4%	4%	5%	5%
Investment income as % of beginning b		(1%)	(0%)	(1%)	(1%)	(1%)	(4%)	(4%)	(4%)	(4%)
Other gain / (loss) as of sales	1%	(10%)	(9%)	(0%)	2%	(0%)	(1%)	(1%)	(1%)	(1%)
Total Operating Expenses Ratio	(26%)	(33%)	(50%)	(30%)	(28%)	(26%)	(16%)	(9%)	(6%)	(5%)
Operating profit	653	493	199	1,046	1,971	2,936	7,114	15,699	27,499	39,971
Operating profit margin	25.5%	13.4%	8.7%	28.8%	36.1%	41.6%	50.7%	56.2%	57.6%	57.5%
Interest Income (Expense)	72	81	179	230	269	511	699	872	1,204	1,718
Income (Loss) from Affiliates	(6)	(4)	(4)	(7)	(15)	(13)	(26)	(26)	(26)	(26)
Non-operating activities	196	843	98	143	95	939	--	--	--	--
Pretax profit	915	1,414	472	1,413	2,321	4,374	7,786	16,545	28,677	41,662
Taxes (enter expense as -)	(86)	(114)	(21)	(72)	(191)	(333)	(934)	(2,151)	(4,015)	(6,249)
Effective tax rate	(9.4%)	(8.1%)	(4.5%)	(5.1%)	(8.2%)	(7.6%)	(12.0%)	(13.0%)	(14.0%)	(15.0%)
Net Profit (incl. minority interest)	829	1,299	451	1,341	2,130	4,040	6,852	14,394	24,662	35,413
Loss/(gain) attributable to the noncontrolli	--	(0)	0	(71)	(106)	(109)	(106)	(106)	(106)	(106)
Minority interest as % of net profit	-	(0.0%)	0.1%	(5.3%)	(5.0%)	(2.7%)	(1.5%)	(0.7%)	(0.4%)	(0.3%)
Profit attributable to equity owners of	829	1,299	451	1,412	2,236	4,149	6,958	14,500	24,768	35,519
Net profit margin%	32.4%	35.4%	19.7%	38.8%	41.0%	58.8%	49.6%	51.9%	51.9%	51.1%
Net Profit YoY		57%	(65%)	213%	58%	86%	68%	108%	71%	43%
Extraordinary gain/ (loss)	--	--	--	--	--	--	--	--	--	--
Basic EPS	0.73	1.15	0.40	1.25	1.97	3.44	5.69	11.74	20.03	28.44
Diluted EPS	0.73	1.15	0.40	1.25	1.96	3.44	5.69	11.74	20.04	28.41
Basic EPS YoY		58%	(65%)	213%	58%	75%	65%	106%	71%	42%
Dividends	339	339	341	338	670	1,675	3,131	6,525	12,384	17,759
DPS	0.30	0.30	0.30	0.39	0.59	1.39	2.56	5.28	10.02	14.22
% payout ratio	41%	26%	75%	31%	30%	40%	45%	45%	50%	50%

Source: Company, Bernstein analysis and estimates

EXHIBIT 19: **Montage Balance sheet**

BALANCE SHEET	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Assets										
Cash & equivalents	5,087	5,834	5,666	6,699	8,471	16,357	19,630	26,631	37,593	53,767
Financial assets held for trading	414	1,610	1,703	1,783	823	2,917	2,917	2,917	2,917	2,917
Accounts & other receivables, prepaym	172	322	294	388	568	613	1,225	2,445	4,184	6,095
Inventories	375	738	482	352	896	826	2,238	4,581	8,166	12,344
Other current assets	1,375	132	150	239	276	530	637	753	890	1,048
Total current assets	7,423	8,636	8,296	9,461	11,034	21,243	26,647	37,328	53,751	76,172
Total PP&E, intangible assets, ROU an	1,178	779	1,094	1,445	1,454	1,692	1,480	1,364	1,311	1,301
Long-term equity investment	290	1,149	1,207	1,226	1,080	660	660	660	660	660
Goodwill	--	--	--	--	--	--	--	--	--	--
Other non-current assets	68	122	101	86	180	306	306	306	306	306
Total non-current assets	1,536	2,050	2,401	2,758	2,714	2,659	2,447	2,330	2,277	2,268
Total assets	8,959	10,686	10,698	12,219	13,748	23,902	29,094	39,658	56,028	78,439
Liabilities and Shareholder's Equity										
Accounts payables and other payables	424	605	376	642	755	1,222	2,515	5,149	9,179	13,874
Short-term borrowings	13	12	13	16	15	21	21	21	21	21
Contract Liabilities	--	--	--	--	1	3	3	3	3	3
Other current liabilities	--	23	2	22	0	0	0	1	1	1
Total Current Liabilities	437	639	391	680	770	1,246	2,539	5,173	9,203	13,899
Long-term borrowings	20	9	2	29	19	122	122	122	122	122
Other non-current liabilities	111	111	98	114	88	124	124	124	124	124
Total Non Current Liabilities	131	119	100	143	107	245	245	245	245	245
Total liabilities	568	759	491	822	877	1,491	2,785	5,419	9,449	14,145
Capital Reserve & Additional Add-in Ca	5,866	1,134	1,139	1,145	6,118	14,507	14,507	14,245	14,245	13,983
Less: Treasury Stock	(300)	(300)	(300)	(428)	(848)	(847)	(847)	(586)	(586)	(324)
Retained earnings / (accumulated defic	2,646	3,622	3,732	4,805	6,371	8,832	12,659	20,634	33,018	50,777
Other comprehensive income / (loss)	--	--	--	--	--	--	--	--	--	--
Other equity tools	179	5,456	5,621	5,881	1,283	51	229	290	352	413
Total Shareholders Equity	8,391	9,912	10,191	11,403	12,924	22,542	26,547	34,583	47,029	64,850
Minority shareholders' interest	--	15	15	(7)	(53)	(132)	(238)	(344)	(450)	(556)
Total Equity	8,391	9,927	10,207	11,397	12,871	22,410	26,309	34,240	46,579	64,294
Total Liabilities & Equity	8,959	10,686	10,698	12,219	13,748	23,902	29,094	39,658	56,028	78,439

Source: Company, Bernstein analysis and estimates

EXHIBIT 20: **Montage cash flow statement**

CASH FLOW STATEMENT	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Net Income	829	1,299	451	1,341	2,130	3,699	6,958	14,500	24,768	35,519
Depreciation and Amortization	59	52	105	141	120	180	655	587	551	538
SBC & Others	243	(271)	193	(23)	361	89	178	62	62	62
Change in net working capital	(155)	(392)	(18)	162	(668)	152	(837)	(1,046)	(1,431)	(1,551)
Cash from operating activities	975	689	731	1,691	2,048	4,121	6,954	14,102	23,951	34,567
Net capex	(153)	(226)	(424)	(379)	(266)	(418)	(444)	(470)	(498)	(528)
Cash paid for equity investment	--	--	--	--	--	--	--	--	--	--
Other investments	(72)	290	(150)	(64)	1,090	(1,379)	--	--	--	--
Cash from investing activities	(226)	63	(574)	(443)	824	(1,815)	(444)	(470)	(498)	(528)
Change in debt	--	(13)	(16)	(21)	--	113	--	--	--	--
Change in equity	(270)	82	(15)	83	(359)	7,110	--	(262)	--	(262)
Cash Paid for Distribution of Dividends, P	(339)	(339)	(341)	(338)	(670)	(1,675)	(3,131)	(6,525)	(12,384)	(17,759)
Other Cash Paid Related to Financing Ac	--	--	--	--	--	(52)	(106)	156	(106)	156
Cash from financing activities	(609)	(270)	(372)	(277)	(1,029)	6,404	(3,237)	(6,631)	(12,490)	(17,865)
Exchange Gains (losses) on Net Cash	(61)	268	47	62	(126)	(300)	--	--	--	--
Net change in cash during period	79	750	(168)	1,033	1,717	8,410	3,273	7,001	10,962	16,174

Source: Company, Bernstein analysis and estimates

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We set our A-share TP at CNY 400, based on 51x 2BF P/E. For the H shares, we set a target price of HKD 520, implying a 15% premium to our A-share TP (we use the CNY to HKD exchange rate at 1:1.13). The H share premium reflects that global investors favor Montage as a scarce China AI-exposed name without direct geopolitical risks, unlike many other Chinese semiconductor companies that face entity list restrictions or export control headwinds. At our target price, the implied P/E multiple on 2BF EPS for H shares is 59x.

RISKS**Montage Technology**

Key potential risks:

1. A decrease in memory and AIDC server demand;
2. Intensify competition that leads to share loss and lower margin;
3. Failure to introduce new products for the AIDC networking market.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

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Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

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- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
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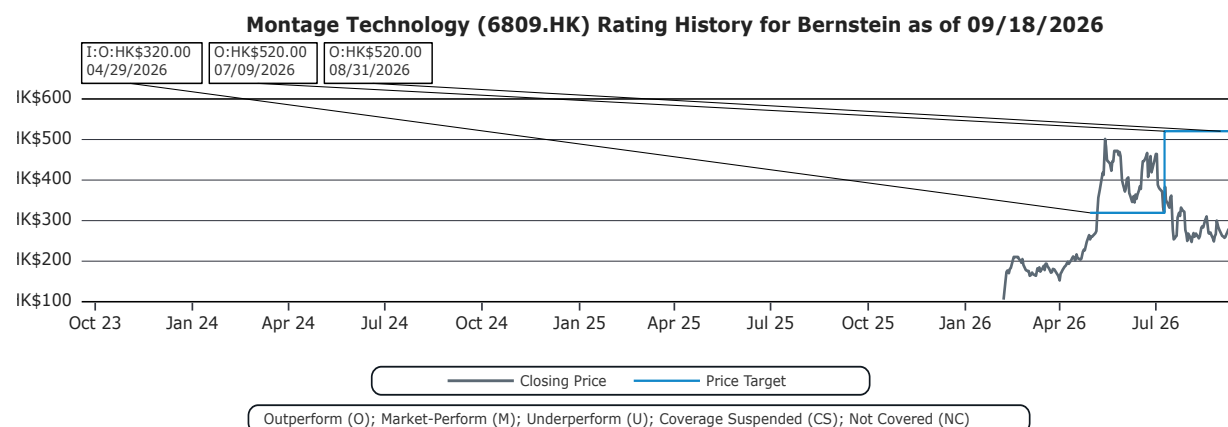
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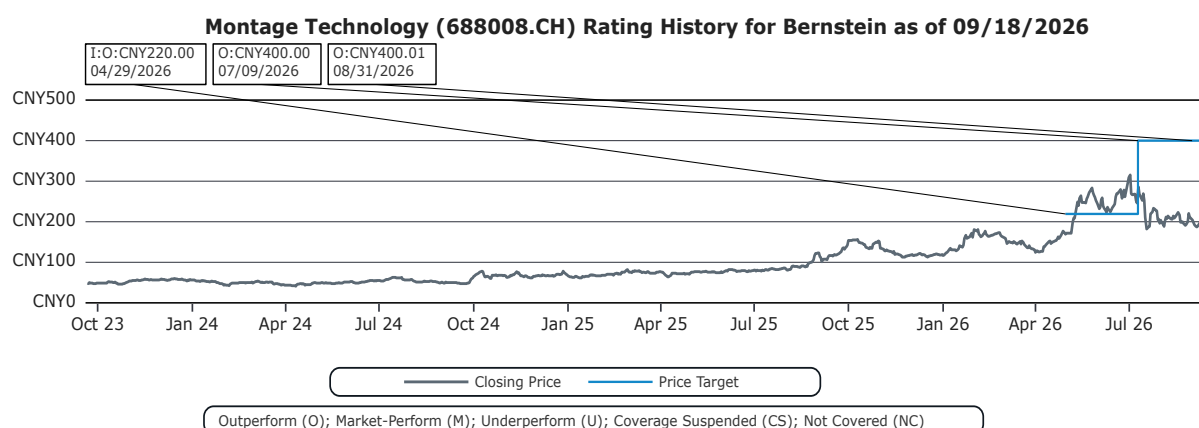
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As of June 30, 2026. All figures are updated quarterly.

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